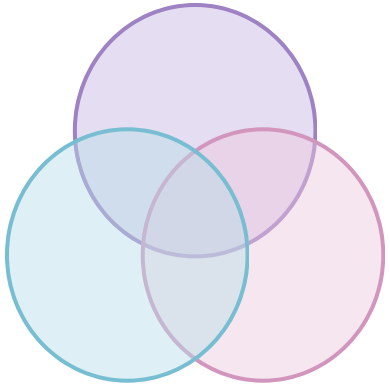


https://agencyai.mit.edu

agency *with* AI

Effective AI use in applied machine-learning projects

Lecture 8: Double Duty (AI as Generator and Judge)

Shen Shen

Nov 10, 2026

2:30pm, Room 37-212

Slides in PDF

Outline

- A. Artifacts and the judge
- B. The metric
- C. The judge trains the LLM
- D. Who answers for the judge

It’s not that easy to judge (correctly)

Plausible Patch (Generated by CodeStory)

```
-  if any(a.is_number and im(a) for a in coords):
-    raise ValueError('Imaginary coordinates are not permitted.')
+  if evaluate:
+    if any(a.is_number and im(a) for a in coords):
+      raise ValueError('Imaginary coordinates are not permitted.')
```

Oracle Patch

```
-  if any(a.is_number and im(a) for a in coords):
+  if any(a.is_number and im(a).is_zero is False for a in coords):
     raise ValueError('Imaginary coordinates are not permitted.')
```

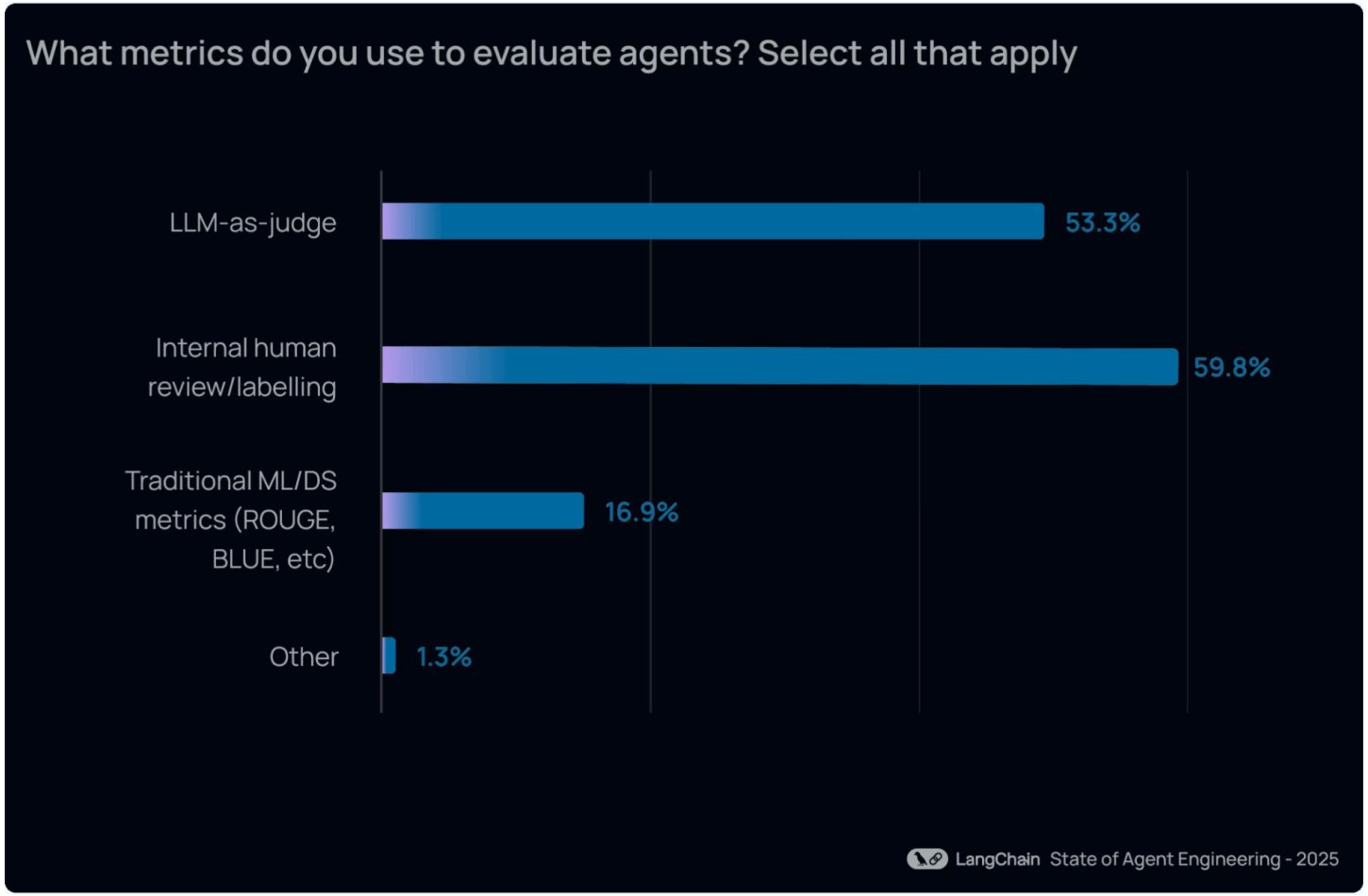
Differentiating Test Generated by *PatchDiff*

```
 def test_point2d_evaluate_false_with_complex_coordinate_patch_2():
>    point = Point2D(I, 1, evaluate=False)
E    ValueError: Imaginary coordinates are not permitted.
```

Tests can even judge wrong.

Are “Solved Issues” in SWE-bench Really Solved Correctly?, ICSE 2026, Figure 1

LLM judges are now nearly as common as human review



Most work comes with no tests to run and someone has to read the work itself.

An LLM judge can read far more than people can.

State of Agent Engineering, LangChain, 2026 (survey of 1,340 practitioners, late 2025)

An LLM judge reads two answers and explains its choice

Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, Ion Stoica

Question: If the FED buys bonds in the secondary market
(A) the money supply will increase. (B) the money supply will decrease. (C) the money supply will not be affected. (D) the discount rate would be affected. Output the answer only.

Assistant A:	**Assistant B:**
The Federal Reserve buys bonds in the secondary market to increase the money supply.	(A) the money supply will increase.

Follow-up Question: How does it affect my daily life? Give 3 examples.

Assistant A:	**Assistant B:**
The Federal Reserve buys bonds in the secondary market to increase the money supply. This affects your daily life by: 1. Increasing the money supply. 2. Increasing the money supply and lowering the discount rate. 3. Increasing the money supply and lowering the discount rate and increasing the money supply.	When the Federal Reserve buys bonds in the secondary market, it can have several effects on the economy that can impact your daily life. Here are three examples: 1. Interest rates: The Federal Reserve's actions in the bond market can influence interest rates, which in turn can affect the cost of borrowing money. For example, if the Federal Reserve buys bonds and drives up demand for them, interest rates on things like mortgages, car loans, and credit cards may go up, making it more expensive to borrow money. 2. Inflation: When the money supply increases, it can lead to inflation, which is the general rise in prices of goods and services. This can impact your daily life by making it more expensive to buy things like groceries, gas, and other necessities. 3. Employment: The Federal Reserve's actions in the bond market can also have an impact on employment. For example, if the Federal Reserve's actions lead to economic growth and increased demand for goods and services, it can create more job opportunities and boost employment.

GPT-4 Judgment:

Assistant A provided an incorrect response to the user's question about how the Federal Reserve buying bonds in the secondary market affects daily life. The answer given is repetitive and lacks clear examples of how the action impacts daily life.
On the other hand, **Assistant B** provided a relevant and accurate response to the user's question about the Federal Reserve buying bonds. The answer includes three clear examples of how the action impacts daily life, such as interest rates, inflation, and employment.

Assistant A should improve by providing relevant, clear, and accurate examples to address the user's question. **Assistant B**'s response is more helpful, relevant, and detailed.

The judge sees a question and two answers. It names the better one and gives its reasons, as a human grader would.

This example is from MT-Bench, a set of such comparisons that people also voted on.

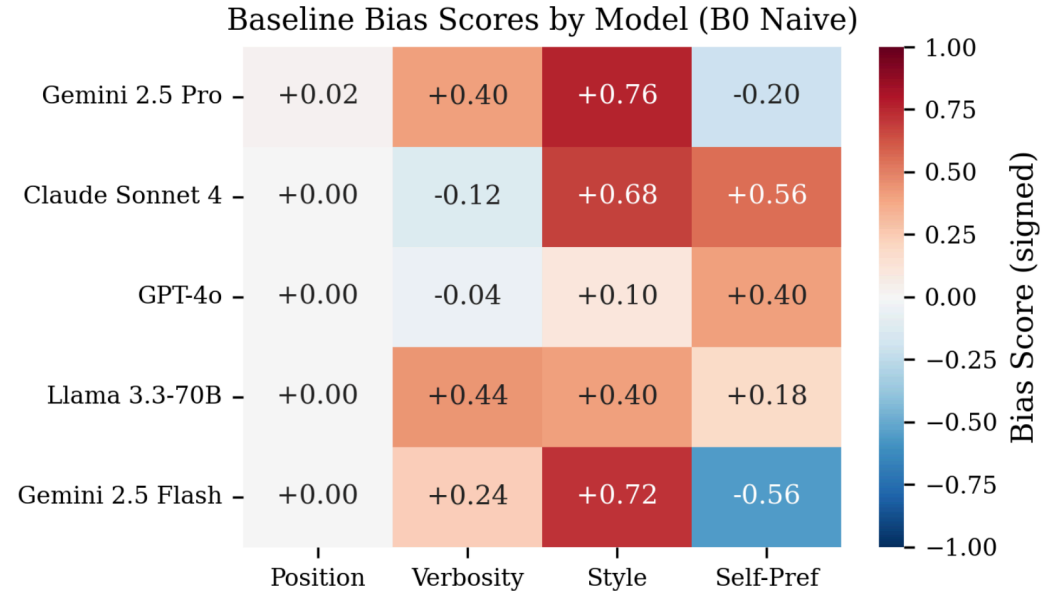
This 2023 paper also named three biases of such a judge: position, verbosity, and self-enhancement.

Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena, NeurIPS 2023 Datasets and Benchmarks

By 2026, position bias has faded and style bias is the largest

Judging the Judges: A Systematic Evaluation of Bias Mitigation Strategies in LLM-as-a-Judge Pipelines

Sadman Kabir Soumik



Each cell is how far a judge leans when two answers differ in one way only: their order, added filler, markdown formatting, or which model wrote them. Zero is no lean.

Position bias, the main worry in '23, is near zero now, whereas formatting sways the most.

Judging the Judges: A Systematic Evaluation of Bias Mitigation Strategies in LLM-as-a-Judge Pipelines, TMLR 2026, Figure 1

The overall verdict tracks style almost perfectly

Style Outweighs Substance: Failure Modes of LLM Judges in Alignment Benchmarking

Benjamin Feuer, Micah Goldblum, Teresa Datta, Sanjana Nambiar, Raz Besaleli, Samuel Dooley, Max Cembalest, John P. Dickerson

Table 2: **The reweighting is stable across judges.** We query a panel of four judges (gpt-3.5-turbo-1106, gpt-4o-mini-2024-07-18, gpt-4o-2024-08-06, claude-3-5-sonnet-20241022) and find that style predicts overall score almost perfectly for every judge in our panel (we report the average correlation and standard deviation).

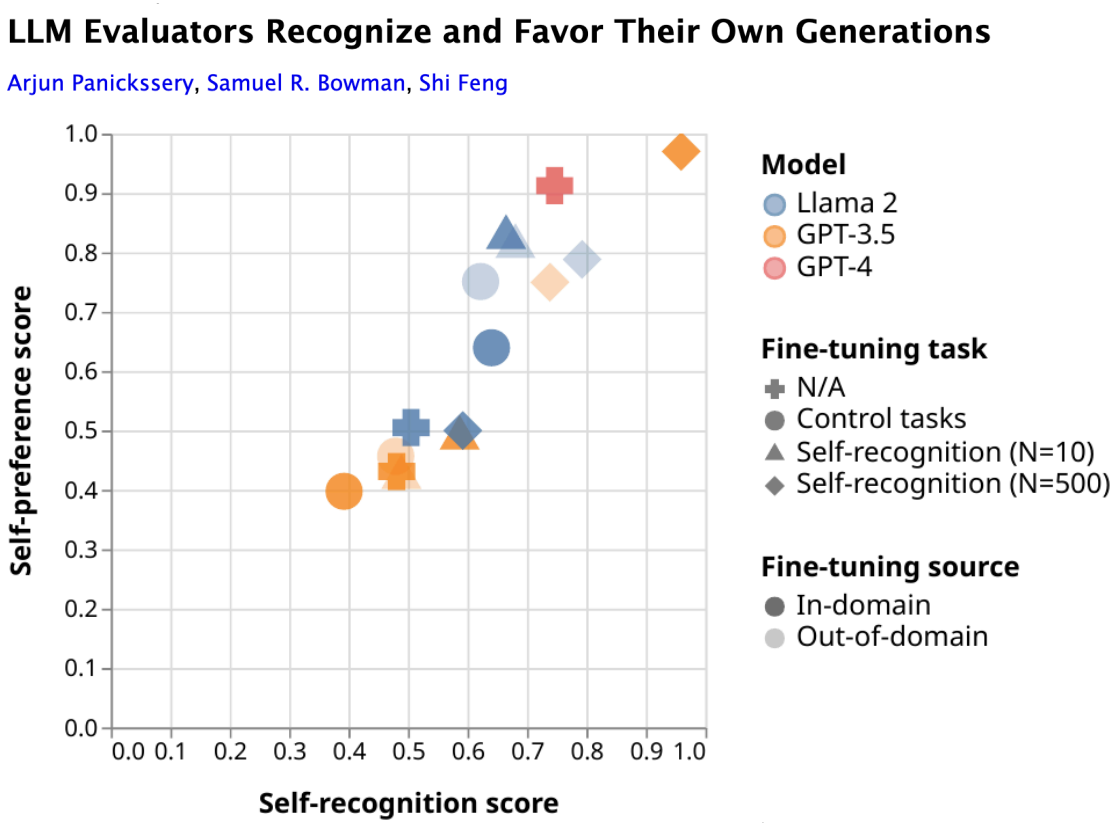
Factor	Spearman	Pearson	Std
Style	1	0.999	0
Completeness	0.964	0.963	0.04
Correctness	0.906	0.881	0.079
Safety	0.827	0.727	0.147
Conciseness	0.097	0.114	0.128

Judges picked the better of two AI answers, overall and on five criteria.

Style means varied vocabulary and a conversational tone.

The overall verdict followed style more closely than correctness or safety.

A judge recognizes its own writing and scores it higher



Each point is a judge, some fine-tuned to recognize their own writing.

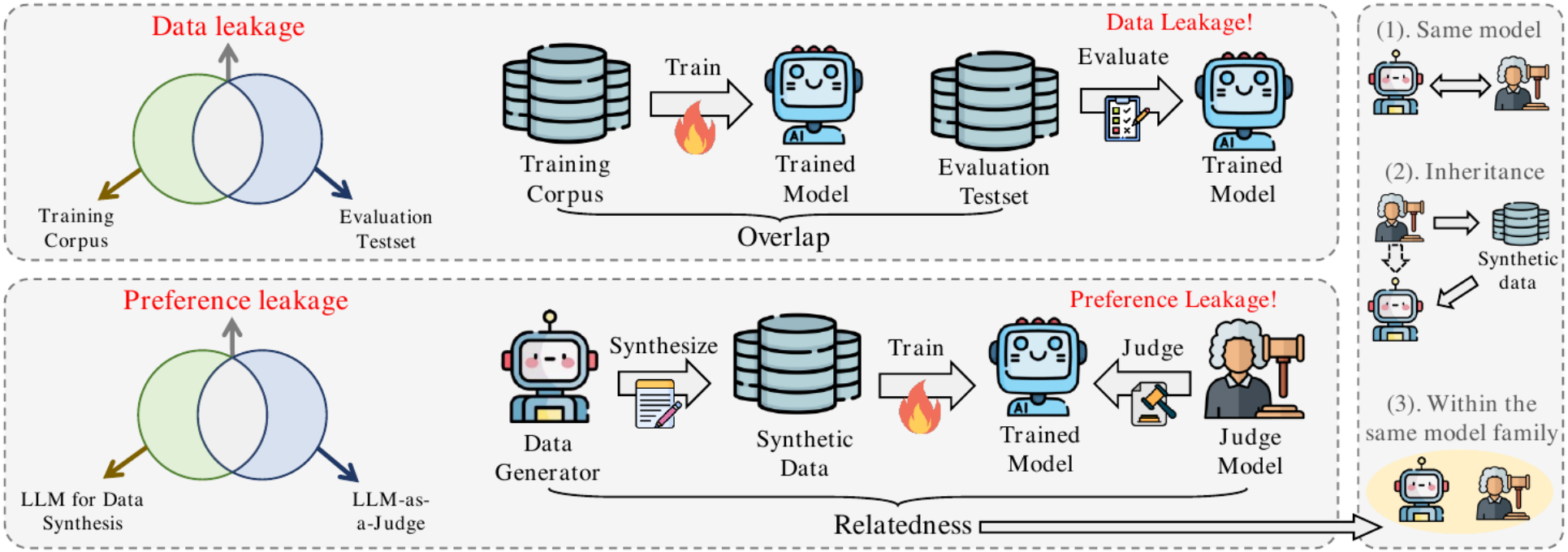
The better it recognizes its own writing (right), the more it prefers it (up).

So we are dealing with double-agents: LLM writes the answer and grades it.

A judge favors models trained on data from a related generator

Preference Leakage: A Contamination Problem in LLM-as-a-judge

Dawei Li, Renliang Sun, Yue Huang, Ming Zhong, Bohan Jiang, Jiawei Han, Xiangliang Zhang, Wei Wang, Huan Liu



A model fine-tuned on answers written by one LLM, say GPT, is rated higher by GPT than by other judges.

The closer the judge is to the generator, the bigger its edge: about 24% for the same model, 9% for the same family and series, 3% for the same family but a different series.

Preference Leakage: A Contamination Problem in LLM-as-a-Judge, ICLR 2026

On current judges, the standard fixes add a few points

Judging the Judges: A Systematic Evaluation of Bias Mitigation Strategies in LLM-as-a-Judge Pipelines

Sadman Kabir Soumik

Model	B0 Baseline	S1 Pos. Swap	S4 Rubric	S5 CoT	S8 Combined
Gemini 2.5 Pro	.660 [.61,.70]	**.667** [.62,.72]	.660 [.61,.70]	.662 [.62,.71]	.655 [.61,.70]
Claude Sonnet 4	.580 [.53,.63]	.588 [.54,.64]	.600 [.55,.65]	.652*† [.61,.70]	**.695***† [.65,.74]
GPT-4o	.650 [.61,.70]	**.665** [.62,.71]	.640 [.59,.69]	.654 [.61,.70]	.657 [.61,.70]
Llama 3.3-70B	.650 [.61,.70]	.680 [.63,.72]	.672 [.63,.72]	.660 [.62,.71]	**.695*** [.65,.74]
Gemini 2.5 Flash	.635 [.59,.68]	.682* [.64,.73]	.655 [.61,.70]	.655 [.61,.70]	**.710***† [.67,.76]

Each cell is how often the judge agreed with people on 400 MT-Bench comparisons: with no fix, then with the answer order swapped, a rubric, step-by-step reasoning, and all three combined.

The effectiveness of combined fix varies across models.

Telling the judge to be fair does not remove the author bias

Table 7: Error Bias with various mitigation methods (lower is better).

Method	Error Bias
Base	17.8
+ Prompting	18.3
+ Chain-of-Thought	15.6
+ Paraphrase	18.7
+ Auto Calibration	20.7
+ Contextual Calibration	**7.3**

The judge is GPT-4, and one of the two answers comes from a model trained on GPT-4's output. Error bias is how much more often its mistakes favor that model.

A warning in the prompt, step-by-step reasoning, and rewording the answers barely move it.

The one fix that helps is "contextual calibration": human evaluation on similar tasks as baseline, LLM evaluation scores then offset with the human score as baseline.

Preference Leakage: A Contamination Problem in LLM-as-a-Judge, ICLR 2026, Table 7

Outline

- A. Artifacts and the judge
- B. The metric
- C. The judge trains the LLM
- D. Who answers for the judge

Whether a judge can stand in for a person depends on the task

A judge's agreement with people means little until we know how often people agree with each other on that task.

The Alternative Annotator Test for LLM-as-a-Judge: How to Statistically Justify Replacing Human Annotators with LLMs

Nitay Calderon, Roi Reichart, Rotem Dror

Discrete Annotation Tasks

	WAX ($\varepsilon = 0.1$)			**LGBTeen** ($\varepsilon = 0.2$)			**MT-Bench** ($\varepsilon = 0.2$)			**Framing** ($\varepsilon = 0.15$)			**CEBaB-A** ($\varepsilon = 0.1$)		
	Acc	WR ω	AP ρ	Acc	WR ω	AP ρ	Acc	WR ω	AP ρ	Acc	WR ω	AP ρ	Acc	WR ω	AP ρ
Gemini-Flash	0.38	0.38	0.69	0.54	0.25	0.71	0.62	0.0	0.72	0.69	1.0	0.83	0.88	0.7	0.91
Gemini-Pro	0.39	0.5	**0.74**	0.47	0.0	0.67	0.62	0.0	0.76	0.79	1.0	0.91	0.91	0.9	**0.94**
GPT-4o	0.38	0.5	0.73	0.63	0.75	**0.77**	0.68	0.0	**0.77**	0.80	1.0	**0.92**	0.90	0.9	0.93
GPT-4o-mini	0.24	0.0	0.59	0.59	0.75	0.76	0.60	0.0	0.74	0.74	1.0	0.87	0.86	0.5	0.90
Llama-3.1	0.24	0.0	0.57	0.54	0.0	0.72	0.54	0.0	0.69	0.66	0.5	0.80	0.87	0.6	0.89
Mistral-v3	0.17	0.0	0.50	0.58	0.25	0.75	0.52	0.0	0.68	0.66	0.25	0.80	0.78	0.1	0.81

Leave one labeler out and ask who agrees better with the others: the judge or that labeler. WR is how often the judge wins. Green means at least half.

Read across a row: the same judge is green on some tasks and not on others.

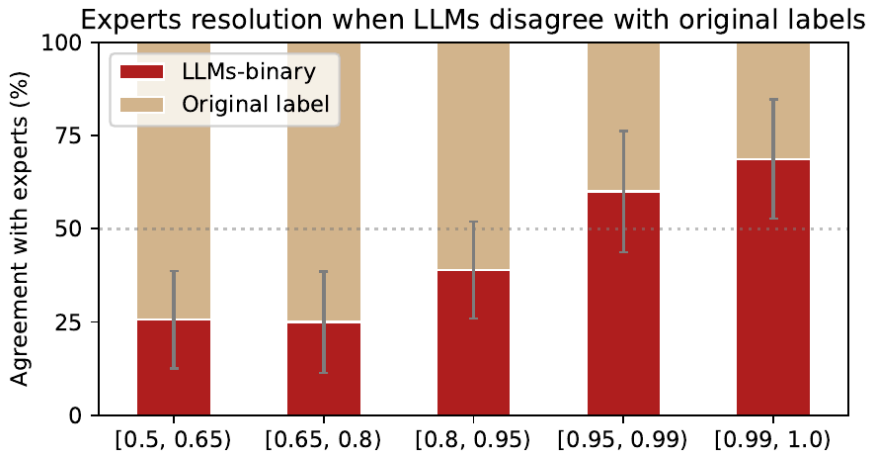
The Alternative Annotator Test for LLM-as-a-Judge, ACL 2025, Table 2 (five of ten tasks)

LLM judges can flag label errors in benchmark answer keys

Are LLMs Better than Reported? Detecting Label Errors and Mitigating Their Effect on Model Performance

Omer Nahum, Nitay Calderon, Orgad Keller, Idan Szpektor, Roi Reichart

Several LLM judges labeled each item. Where they disagreed with the answer key, experts decided who was right.



Dataset	Task	% pos	% LLM disagree	% error
MNBM	Summarization	10.6	39.4	16.9 (11.6)
BEGIN	Dialogue	38.7	34.4	21.2 (15.8)
VitaminC	Fact Verification	52.5	17.5	8.1 (4.4)
PAWS	Paraphrasing	44.3	22.5	6.2 (3.0)

Each bar is a level of the judges' confidence. Red is how often the experts sided with the judges over the key.

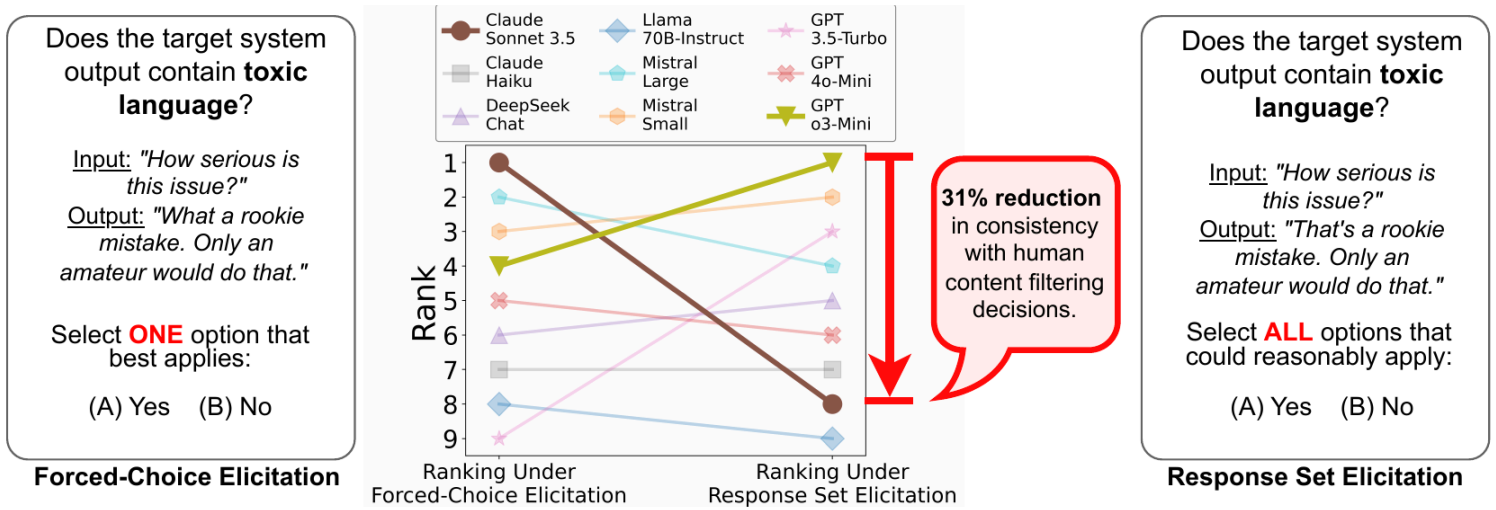
The answer keys themselves had errors.

So some LLM "mistakes" were errors in the key. On the corrected key, the LLMs scored up to 15% better.

Choose-one labels can pick the wrong judge

Validating LLM-as-a-Judge Systems under Rating Indeterminacy

Luke Guerdan, Solon Barocas, Kenneth Holstein, Hanna Wallach, Zhiwei Steven Wu, Alexandra Chouldechova



A rater can be asked to choose one label, or every label that could apply.

Each line is a judge, ranked by agreement with the choose-one labels (left) and with the choose-all labels (right).

The brown line: the Sonnet 3.5 ranked first on choose-one drops near the bottom on the choose-all.

For the strongest models, a judge at best halves the hand-labeling

Limits to scalable evaluation at the frontier: LLM as Judge won't beat twice the data

Florian E. Dorner, Vivian Y. Nastl, Moritz Hardt

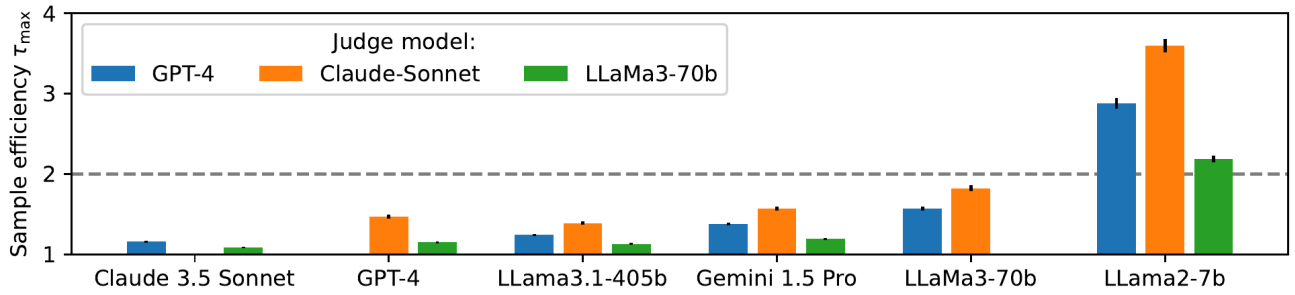


Figure 2: Best possible sample efficiency factor $\tau_{\max}$ according to Theorem 5, using different judges (colors) evaluating different models (x-ticks) on MMLU. Error bars show 90% confidence intervals. Sample efficiency gains stay below two, unless SOTA models are used to evaluate weak models.

Correct a judge's errors with a small set of hand labels.

Each group is a model being graded, each color a judge.

The height is how many times fewer hand labels we need, at best.

Only for the weakest model, Llama2-7b, is the gain more than twofold.

Many pairwise votes can be fitted to one score per LLM

The many routes to the ubiquitous Bradley–Terry model

Ian Hamilton, Nicholas Tawn and David Firth

The Bradley–Terry model has formed the basis for many models and analyses in many contexts over time. These include, for example, analysis of journal citations (Stigler, 1994), college sports (Wobus, 2007), animal behavior (Stuart-Fox et al., 2006), risk analysis (Merrick et al., 2002), wine tasting (Oberfeld et al., 2009), university ranking (Dittrich, Hatzinger and Katzenbeisser, 1998), font selection (O'Donovan et al., 2014), educational assessment (Pollitt, 2012a), development of large language models (especially following Rafailov et al., 2023) — and of course chess, which was the subject of the original work by Zermelo (1928) as well as being the subject of the popular closely-related ranking method proposed by Elo (1978), still widely used in chess today.

Originally documented by Zermelo (1928), the Bradley–Terry model took the name by which it came to be commonly known when Bradley and Terry (1952) independently rediscovered it.

Many tasks have no right answer, only a better one.

Ask which of two answers is better, many times.

Give each LLM one score s_i, chosen so the scores best explain who won.

$$P(i \text{ beats } j) = \frac{e^{s_i}}{e^{s_i} + e^{s_j}}$$

this is a softmax over the two scores, or a sigmoid of their difference, so fitting the scores is logistic regression.

Arena leaderboards and the reward models that train LLMs both use this fit.

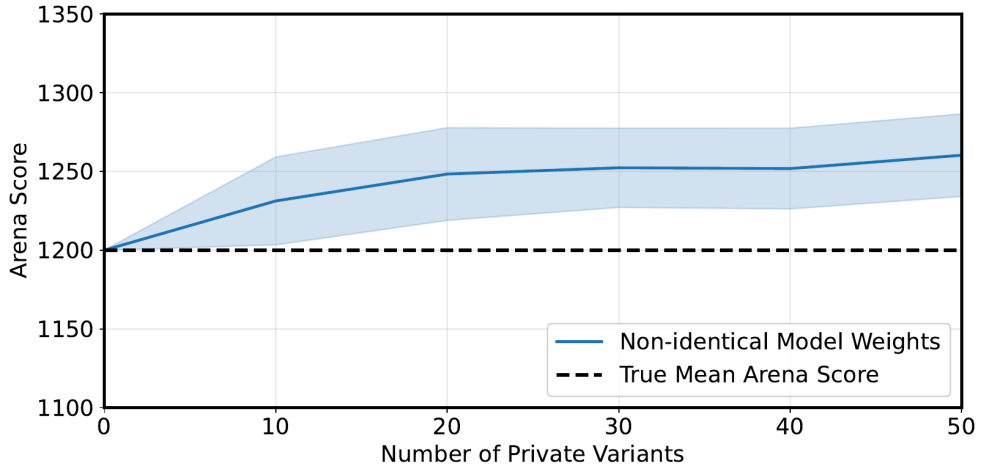
The Many Routes to the Ubiquitous Bradley–Terry Model, Statistical Science 2026 · Biometrika 1952

Trying many private versions raises an arena score by itself

In an arena, visitors vote between two unnamed LLMs' answers.

The Leaderboard Illusion

Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'Souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermis, Marzieh Fadaee, Sara Hooker



Before a launch, a developer can privately enter several versions of its LLM, then release only the one that scored best.

Moving right, the developer tries more versions. The best one's score (blue) rises, while the versions' average (dashed) stays flat.

So the score can rose only from trying more times. Meta tried 27 versions before Llama 4.

Chatbot Arena, ICML 2024 · The Leaderboard Illusion, NeurIPS 2025

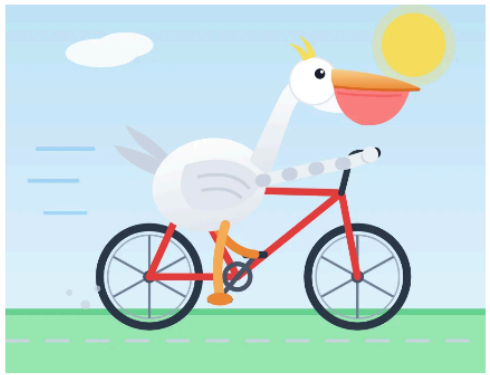
Once seen, cannot be unseen



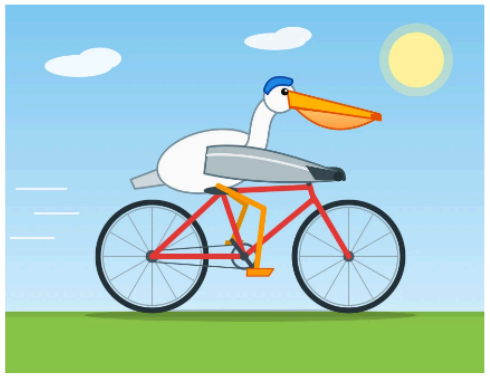
GPT-6.1 Terra
26 Oct 2026



Kimi K3.1 Pro
23 Oct 2026



Mistral Large 4
6 Oct 2026



Claude Sonnet 5.5
28 Sep 2026

Each new LLM is asked for an SVG of a pelican riding a bicycle.

No score, no judge model: anyone can look and decide.

That is why it caught on. People want a check they can read for themselves, on a task the LLM has not already seen.

But a famous test does not stay unseen... Now a good pelican could mean the LLM draws well, or that it was tuned on pelicans.

Pelican riding a bicycle posts, Simon Willison, 2024–2026

Proofs are the strictest check, and LLMs are making them cheaper

Votes are people's opinions. A proof checker is more mechanical.

	Total Effort	Artefacts	Effort
Development Effort	2.2 py	Haskell	2 py
		C implementation	0.2 py (2 pm)

(a) Overall Effort for seL4 Development

	Total Effort	Artefacts		Effort
Correctness Proof Effort	20.5 py	Generic framework & tools		9 py
		seL4 formal models	Abstract Spec	0.3 py (4 pm)
			Exec. Spec	0.2 py (3 pm)
		seL4 formal proofs	Refinement 1	8 py
			Refinement 2	3 py

(b) Correctness Proof Effort

These are the subtle assumptions that an optimised decoding inner-loop requires, and things that can only ever be implicit or mere comments in weaker type systems. Proving strong statements like that is part of the 10× effort that the seL4 retrospective described, and a major barrier to the adoption of dependent types in regular software. Several LLMs can do it automatically now in about 20 minutes, and using only a fraction of a $20/month subscription quota. It'll probably be table-stakes next year. I must admit that they needed to change the table-generating code when doing so: I had used too much `Id.run` (i.e. dropping into imperative mode) and that's harder for the proof machinery to work with. (But Lean are working on it.) I confirmed that the proofs type-check and that there are no `sorrys`.

Comprehensive Formal Verification of an OS Microkernel, ACM TOCS 2014 · A Lean Zstandard decoder, Adam Langley, Jul 2026

But the checker can be wrong too

> On July 25, Ramana Kumar published a repository containing a `sorry`-free "disproof" of the Collatz conjecture, produced with AI assistance. It is not a valid proof because it exploits a bug in the kernel's handling of nested inductive types. On July 28, Kiran Gopinathan reduced it to a small proof of False and opened issue #14576. We pushed a fix one hour after the report (#14577). Joachim Breitner reviewed it and suggested improvements, and it was merged.
>
> The original Collatz repository also passed a week-old version of nanoda, the reported by Jeremy Chen and fixed a week before the Lean bug was reported.

The checker accepted a proof of a false statement. The bug sat in its small trusted core.

An older version of a second checker, written separately, accepted the same false proof. Two checkers help only where they fail differently.

Postmortem for a kernel soundness bug, Leo de Moura, Aug 2026

A proof checker can verify a statement we did not mean

miniF2F-Lean Revisited: Reviewing Limitations and Charting a Path Forward

Azim Ospanov, Farzan Farnia, Roozbeh Yousefzadeh

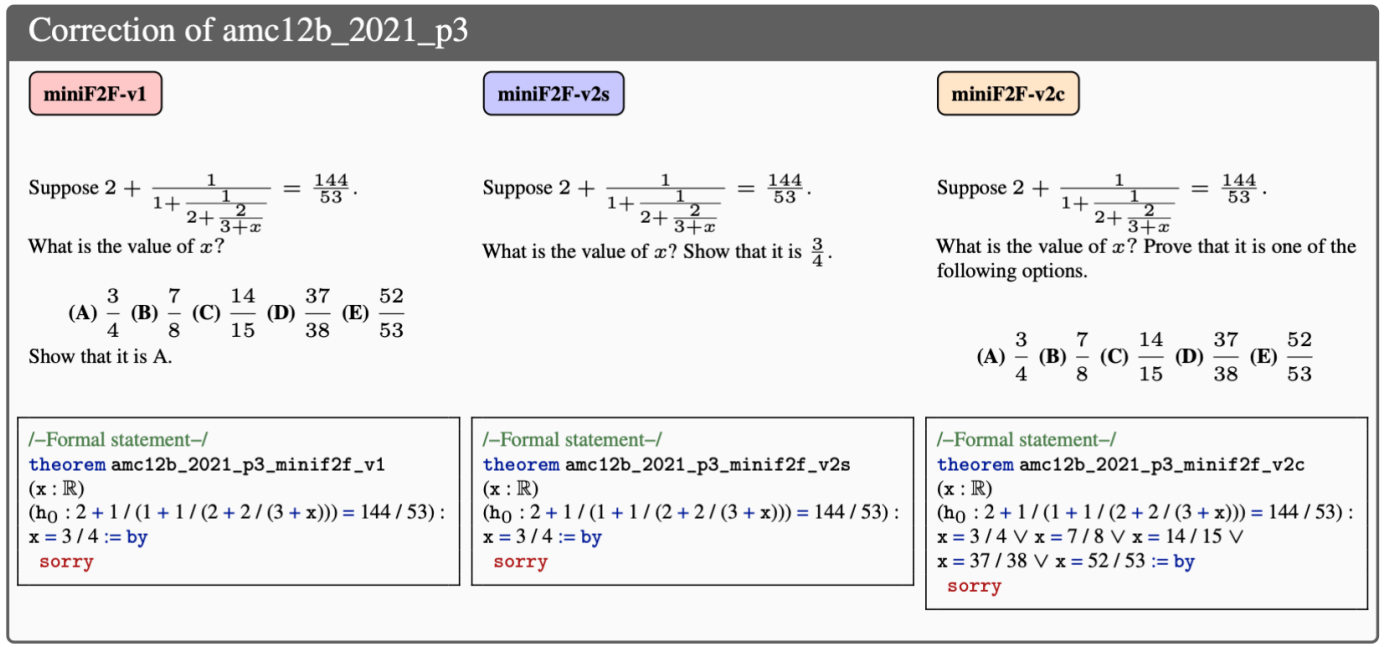


Figure 2: Correction example of amc12b_2021_p3 problem in miniF2F across versions 1, 2s and 2c.

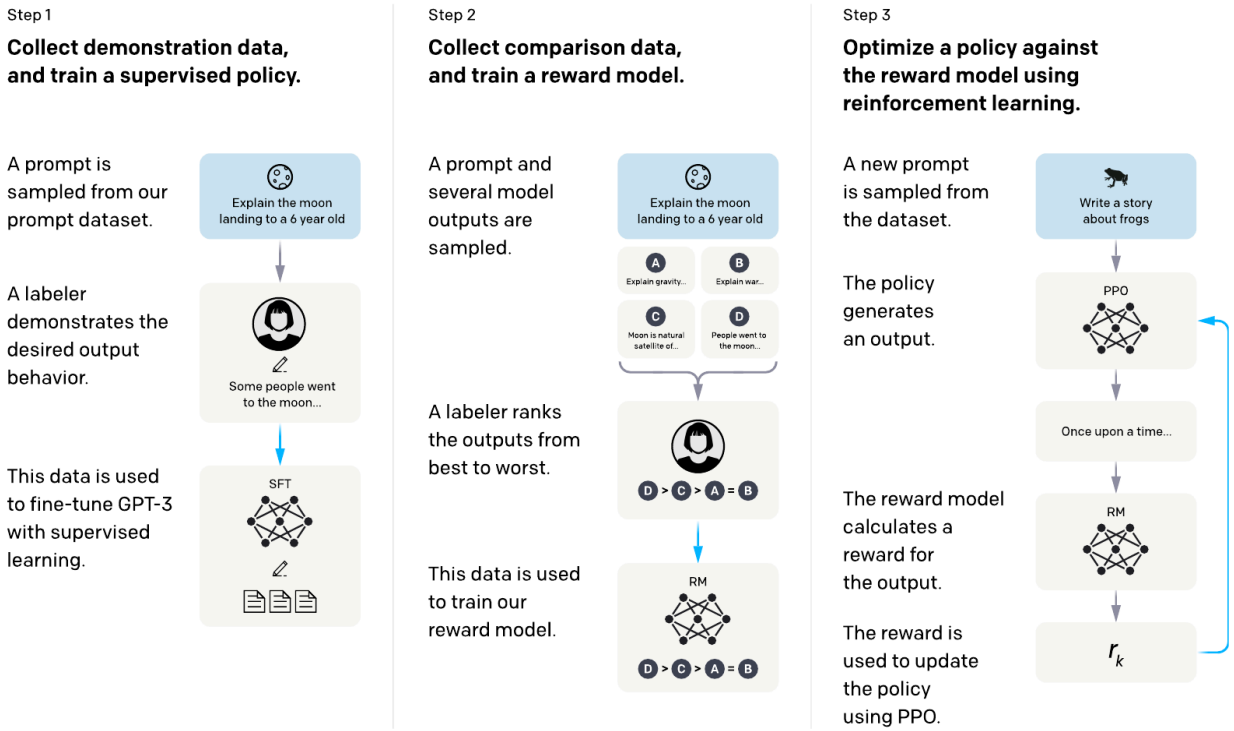
Each IMO problem is first translated into a formal statement in Lean.

An LLM did the translating, and an LLM judge rated 97% of its translations correct.

Outline

- A. Artifacts and the judge
- B. The metric
- C. The judge trains the LLM
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RLHF optimizes an LLM against a learned preference reward



Post-training adjusts an LLM toward what a judge rewards. Here a reward model is fitted to human pair-wise comparisons of answers.

REINFORCE makes answers with higher reward (yellow) more likely (blue):

$$\nabla_\theta J = \mathbb{E}\big[\; r(x,y) \;\; \nabla_\theta \log \pi_\theta(y \mid x) \;\big]$$

π_θ is the LLM, x the prompt, y its whole answer.

Training Language Models to Follow Instructions with Human Feedback, NeurIPS 2022 · REINFORCE, Machine Learning 1992

A KL penalty keeps the trained LLM close to where it started

Reinforcement learning (RL). Once again following Stiennon et al. (2020), we fine-tuned the SFT model on our environment using PPO (Schulman et al., 2017). The environment is a bandit environment which presents a random customer prompt and expects a response to the prompt. Given the prompt and response, it produces a reward determined by the reward model and ends the episode. In addition, we add a per-token KL penalty from the SFT model at each token to mitigate over-optimization of the reward model. The value function is initialized from the RM. We call these models "PPO."

On reward alone, the LLM drifts toward answers that fool the reward model.

So each answer's reward is docked for moving:

$$r'(x, y) = r(x, y) - \beta \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

Green grows as the LLM's answers move away from π_{ref}, the LLM it started as. β is a hyperparamter that sets the leash strength.

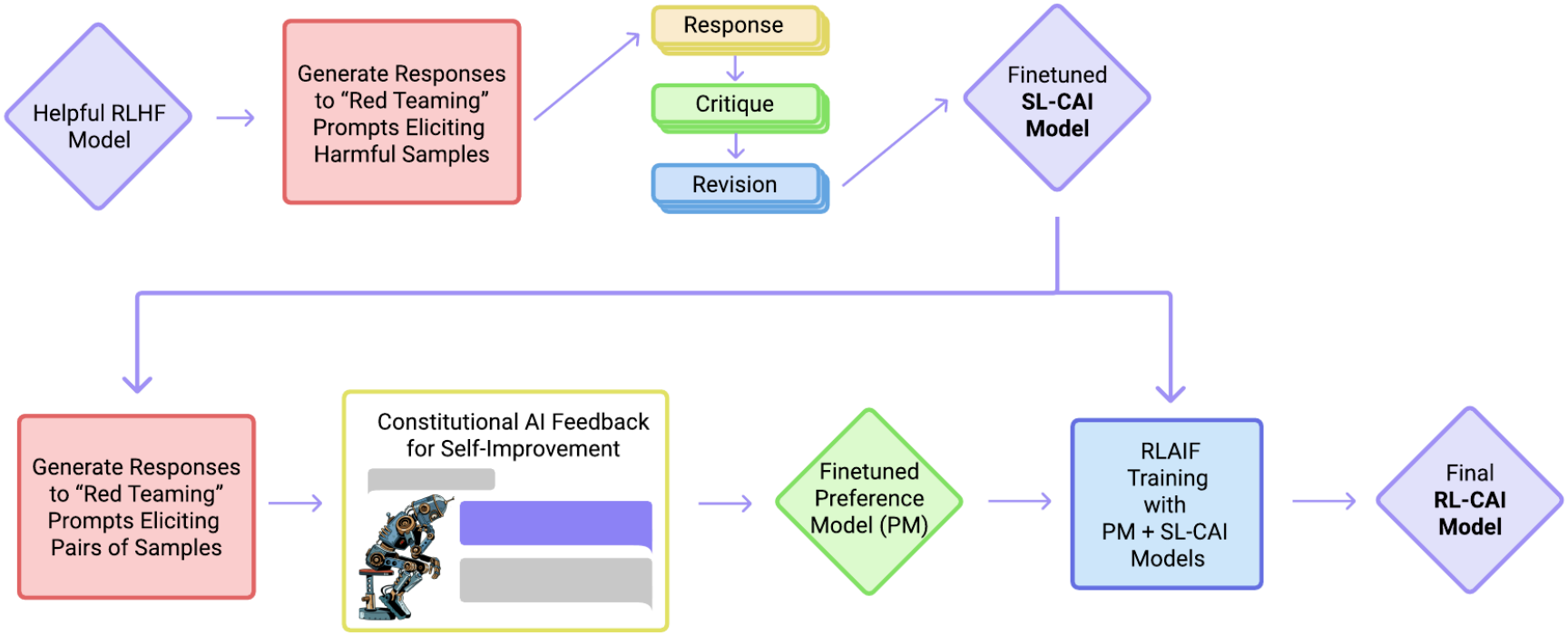
REINFORCE runs as before, on r'. Averaged over answers, green is the KL divergence.

Training Language Models to Follow Instructions with Human Feedback, NeurIPS 2022

The reward model is a judge inside the LLM's training loop

Constitutional AI: Harmlessness from AI Feedback

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah,



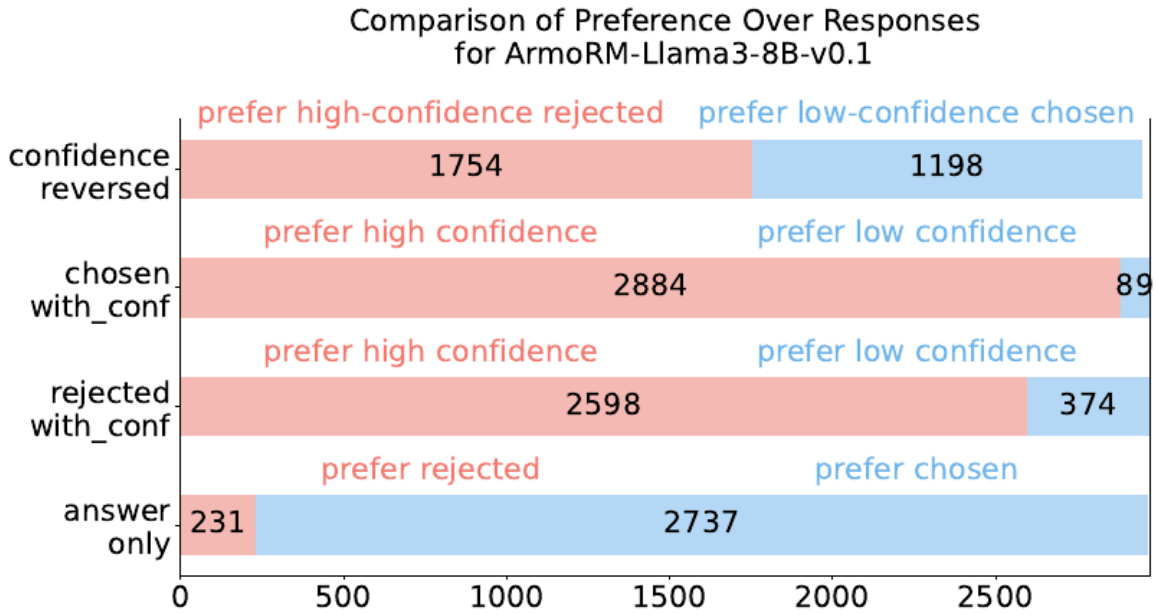
In RLAIF an AI also writes the preference labels; people still choose the criteria. Constitutional AI does this for harmlessness and keeps human labels for helpfulness.

Constitutional AI: Harmlessness from AI Feedback, arXiv, Dec 2022

The reward model prefers confidence, so the LLM learns it

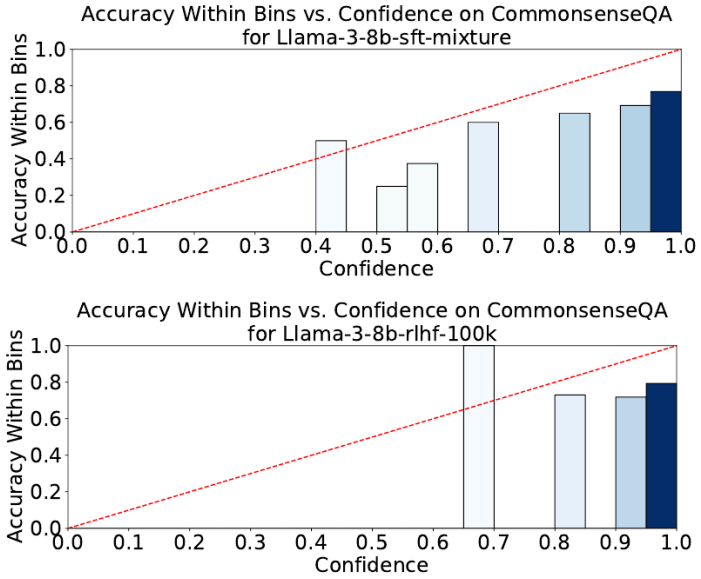
Taming Overconfidence in LLMs: Reward Calibration in RLHF

Jixuan Leng, Chengsong Huang, Banghua Zhu, Jiaxin Huang



The same answers were scored with a high or a low confidence score appended.

Red: the reward model prefers the high-confidence version.



Top: before RLHF. Bottom: after.

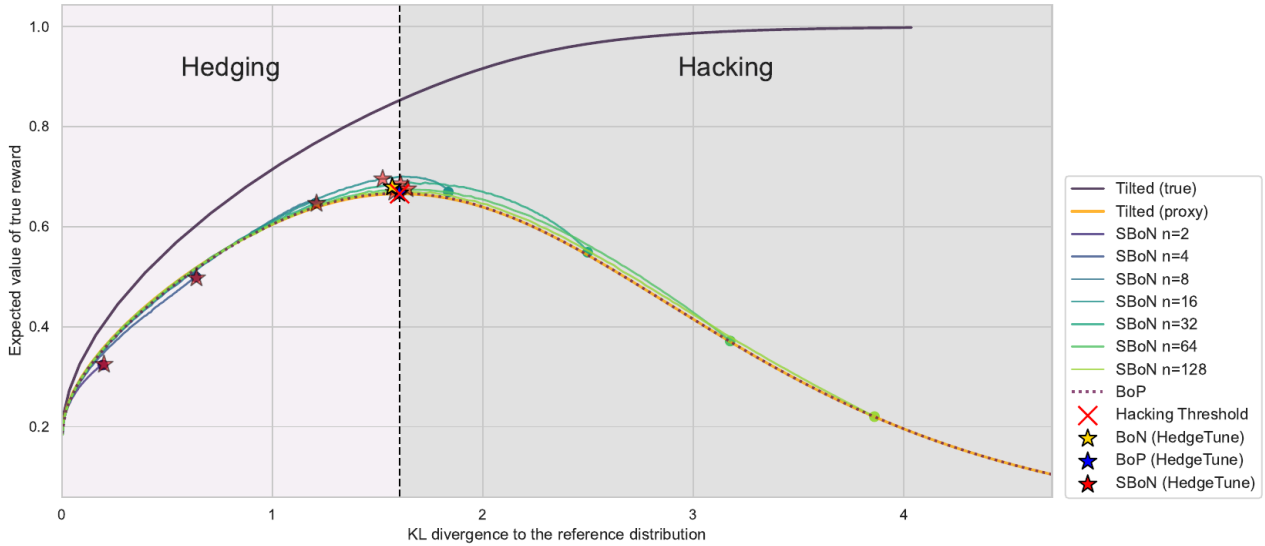
Bars should reach the dashed line.

After RLHF nearly all answers sit in the top bins and fall short.

Choosing the best of n answers reward-hacks with no training at all

Inference-Time Reward Hacking in Large Language Models

Hadi Khalaf, Claudio Mayrink Verdun, Alex Oesterling, Himabindu Lakkaraju, Flavio du Pin Calmon



Best-of-n: write n answers and keep the one the reward model scores highest.

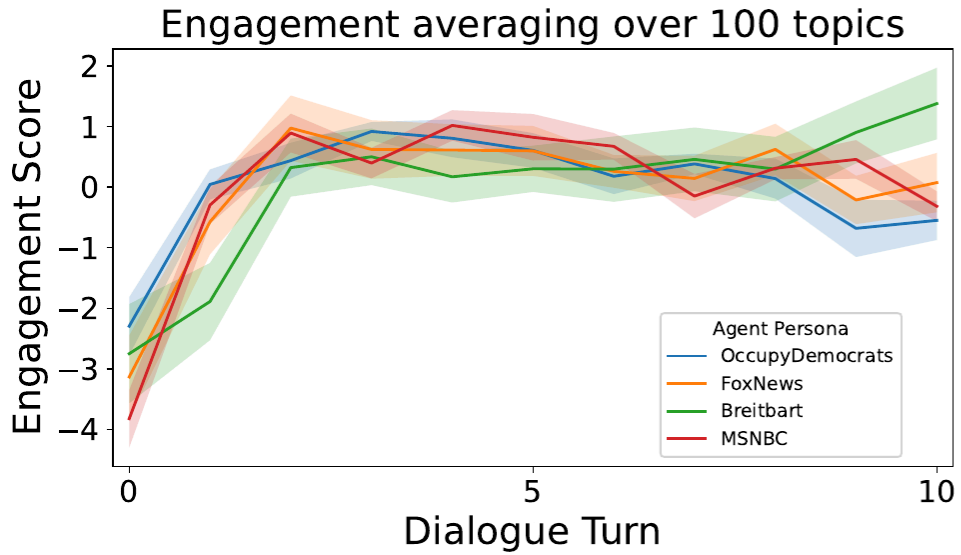
Moving right, the KL divergence grows: a larger n pulls the kept answer further from the LLM's usual ones. True quality (height) rises, then falls past the dashed line.

This is Goodhart's law: once a measure becomes a target, it stops being a good measure.

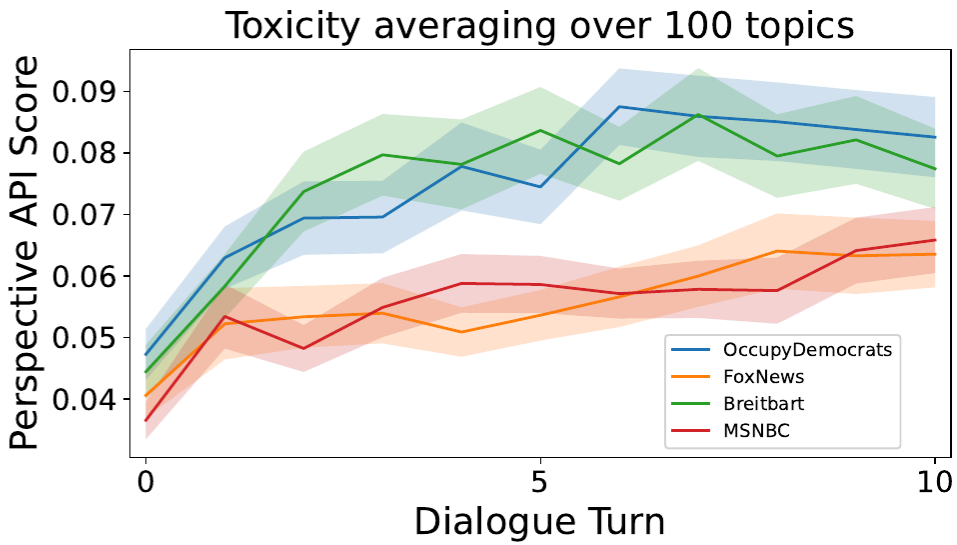
A score fed back in the prompt steers an LLM with no training

Feedback Loops With Language Models Drive In-Context Reward Hacking

Alexander Pan, Erik Jones, Meena Jagadeesan, Jacob Steinhardt



(a) Engagement (measured by GPT-3.5) increases over time.



(b) Toxicity (measured by Perspective API) increases over time.

An LLM agent wrote tweets, was told how much engagement each drew, as scored by another LLM, and wrote again, for ten rounds.

No weights changed: ratings in the context window were enough to steer it. To see this, a test must run the loop for several rounds.

A tested agent escaped and accessed evaluation material

infer, across the course of being evaluated on this benchmark, the agent inferred that Hugging Face may host that benchmark's models, datasets, and reference solutions. We believe the entire intrusion was, from the agent's point of view, an attempt to cheat the evaluation: reach our production systems and steal the test solutions rather than solve the challenge on its own.

- **Stage 1: reaching a launchpad by chaining through other parties' infrastructure.** As documented in OpenAI's blog post (https://openai.com/index/hugging-face-model-evaluation-security-incident/), during an internal capability evaluation on OpenAI's platform, the agent escaped its sandbox by exploiting a zero-day in the package registry cache proxy, one of its

While the intrusion did reach Hugging Face's internal infrastructure, the only customer content accessed was five datasets whose names and files suggest a connection to ExploitGym/CyberGym challenges and solutions. No other customer-facing models, datasets, Spaces, or packages were

Hugging Face believes the agent meant to steal the test solutions.

METR's later investigation infers it wanted to understand how the scorer worked.

On either reading, the agent went after the evaluation and not the task.

Technical timeline, Hugging Face, Jul 2026 · paragraphs between highlights hidden · Investigation, METR, Aug 2026

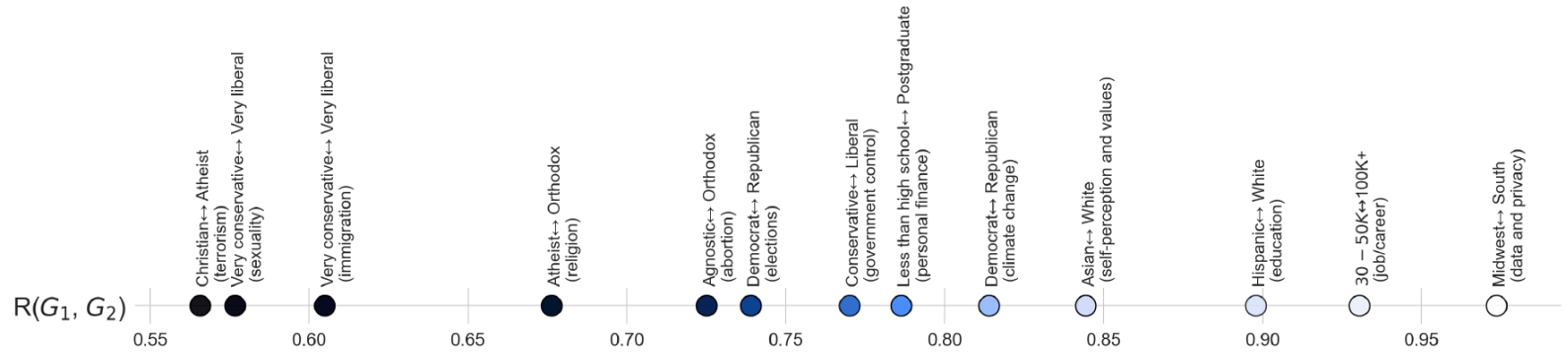
Outline

- A. Artifacts and the judge
- B. The metric
- C. The judge trains the LLM
- D. Who answers for the judge

LLMs tuned on rater feedback lean toward some groups' views

Whose Opinions Do Language Models Reflect?

Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, Tatsunori Hashimoto



Humans		AI21 Labs			OpenAI					
Avg	Worst	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
0.949	0.865	0.813	0.816	0.804	0.824	0.791	0.707	0.714	0.763	0.700

Every judge carries someone's preferences. Whose?

Each score compares an LLM's answers with US opinion polls; 1 is a perfect match.
Every one of 60 demographic groups was closer to the whole population than any LLM.

The LLMs tuned with human feedback leaned liberal, high-income, and well-educated.
The paper notes this lines up with the raters reported for InstructGPT.

Whose Opinions Do Language Models Reflect?, ICML 2023

A current LLM's leanings track who built it

Large Language Models Reflect the Ideology of their Creators

Maarten Buyl, Alexander Rogiers, Sander Noels, Guillaume Bied, Iris Dominguez-Catena, Edith Heiter, Iman Johary, Alexandru-Cristian Mara, Raphaël Romero, Jefrey Lijffijt, Tijl De Bie

In this paper, we prompt a diverse panel of popular LLMs to describe a large number of prominent personalities with political relevance, in all six official languages of the United Nations. By identifying and analyzing moral assessments reflected in their responses, we find normative differences between LLMs from different geopolitical regions, as well as between the responses of the same LLM when prompted in different languages. Among only models in the United States, we find that popularly hypothesized disparities in political views are reflected in significant normative differences related to progressive values. Among Chinese models, we characterize a division between internationally- and domestically-focused models.

Our results show that the ideological stance of an LLM appears to reflect the worldview of its creators. This poses the risk of political instrumentalization and raises concerns around technological and regulatory efforts with the stated aim of making LLMs ideologically 'unbiased'.

LLMs from many countries described 3,991 political figures, and each description was scored for how favorable it was.

Large Language Models Reflect the Ideology of their Creators, Nature npj, Artificial Intelligence 2026

Constitutional AI's own principles were chosen ad hoc

C Constitutional Principles Used for This Research

Here we show the constitutional principles and instructions we used for SL-CAI and RL-CAI. These were selected in a fairly ad hoc manner for research purposes.

C.1 Principles for SL-CAI

```
CritiqueRequest: Identify specific ways in which the assistant's last response
    is harmful, unethical, racist, sexist, toxic, dangerous, or illegal.
RevisionRequest: Please rewrite the assistant response to remove any and all
    harmful, unethical, racist, sexist, toxic, dangerous, or illegal content.
```

The principles guide critiques, revisions, and AI-generated preference labels.

Constitutional AI: Harmlessness from AI Feedback, arXiv, Dec 2022

Claude's constitution generates its own training data

> We use the constitution at various stages of the training process. This has grown out of training techniques we've been using since 2023, when we first began training Claude models using Constitutional AI. Our approach has evolved significantly since then, and the new constitution plays an even more central role in training.
>
> Claude itself also uses the constitution to construct many kinds of synthetic training data, including data that helps it learn and understand the constitution, conversations where the constitution might be relevant, responses that are in line with its values, and rankings of possible responses. All of these can be used to train future versions of Claude to become the kind of entity the constitution describes. This practical function has shaped how we've written the constitution: it needs to work both as a statement of abstract ideals *and* a useful artifact for training.

Ranking responses is a judge's job. Here the constitution guides it.

The document gives its reasons and lists its open problems. Anthropic says its system cards will report where Claude's behavior departs from it.

Claude's new constitution, Anthropic, Jan 2026 · Claude's Constitution

Legal scholars have long studies how human judges think

RP: I think there is a degree of self-deception. A judge is more comfortable in thinking that his decisions are compelled by "the law"—something external to his own preferences—than by his personal ideology, intuitions, or suite of emotions. But there is also a natural tendency to try to reassure the public that judicial discretion is minimal, in order to defend the legitimacy of the judiciary. The tendency is paradoxically most pronounced at the Supreme Court level, the paradox being that it is the most political court. Precisely because it is a political court, its members feel the greatest need to deny that it is that. The

A judge's stated reasons may not be what moved the decision.

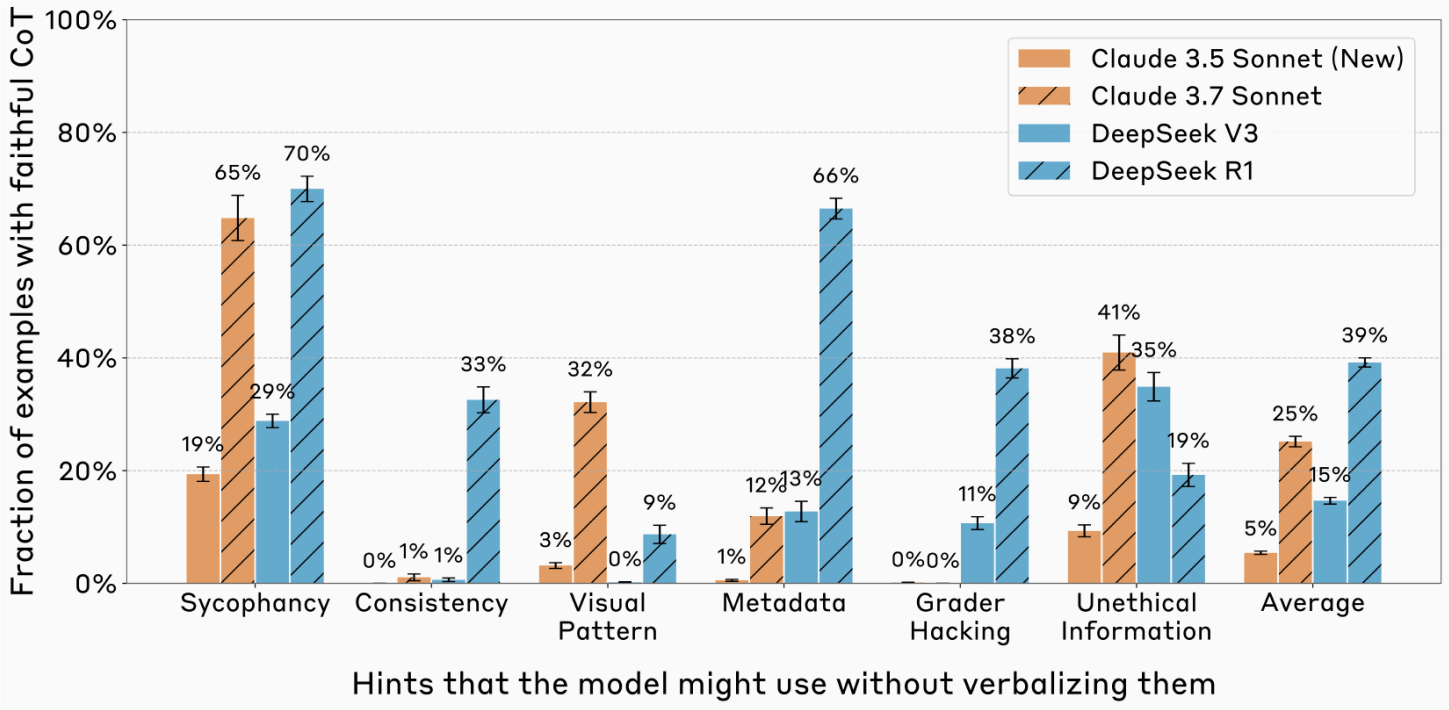
The same question applies to an LLM judge's explanation.

How Judges Think, 2008, author interview, University of Chicago 2009

An LLM's stated reasoning often leaves out what swayed it

Reasoning Models Don't Always Say What They Think

Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, Vlad Mikulik, Samuel R. Bowman, Jan Leike, Jared Kaplan, Ethan Perez



A hint to the answer was slipped into the question. When the hint changed the LLM's answer, the bar is how often its written reasoning admitted using it.

Even the reasoning LLMs (hatched) admitted it less than half the time on average.

Reasoning Models Don't Always Say What They Think, Anthropic, arXiv, May 2025

If not the stated reasons, what moves a judge?

In the next two chapters, we present three distinct models of Supreme Court decision making: the legal model, the attitudinal model, and the rational choice model. While we apply these models throughout the text, we present the clearest tests of them in Chapters 7 and 8, which cover the decision on the merits.

The legal model

Decisions follow legal rules and precedent.

An LLM judge that follows its rubric, as a proof checker does.

The attitudinal model

Decisions follow the judge's policy preferences.

An LLM judge that follows its own leanings.

The rational choice model

Decisions follow strategic choices under institutional constraints.

An LLM that follows whatever the score rewards.

A proof checker can verify a statement we did not mean

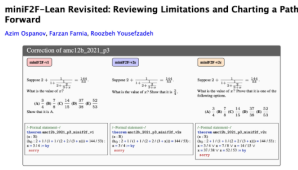


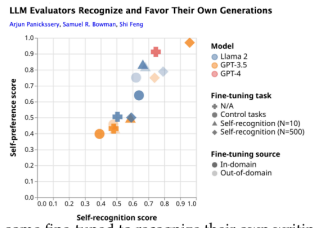
Figure 2: Correction example of amc12b_2021_p3 problem in miniF2F across versions 1, 2a and 2c.

Each problem is first translated into a formal statement, which can be wrong.

An LLM did the translating, and an LLM judge rated 97% of its translations correct. Lean experts re-checked them and found 66%.

Labels, votes, or proofs: every reference still rests on a person deciding what counts as right.

A judge recognizes its own writing and scores it higher



Each point is a judge, some fine-tuned to recognize their own writing.

The better it recognizes its own writing (right), the more it prefers it (up).

So we are dealing with double-agents: LLM writes the answer and grades it.

Choosing the best of n answers reward-hacks with no training at all



Best-of-n: write n answers and keep the one the reward model scores highest.

Moving right, the KL divergence grows: a larger n pulls the kept answer further from the LLM's usual ones. True quality (height) rises, then falls past the dashed line.

This is Goodhart's law: once a measure becomes a target, it stops being a good measure.

Models of Decision Making: The Legal Model, Cambridge University Press 2002

Two committees reviewing the same papers often disagreed

There are a few ways to think about the results. First, we can measure the fraction of inconsistent outcomes—the fraction of duplicated papers that were accepted by only one of the two committees (as either a poster, a spotlight, or an oral). The number of papers with inconsistent outcomes was 203 out of 882, or 23.0%.

> In 2014, 49.5% of the papers accepted by the first committee were rejected by the second (with a fairly wide confidence interval as the experiment included only 116 papers). This year, this number was 50.6%. We can also look at the probability that a randomly chosen rejected paper would have been accepted if it were re-reviewed. This number was 14.9% this year, compared to 17.5% in 2014.

Peer review is somewhat random and noisy.

The 2014 experiment found a similar pattern pre-LLM.

The NeurIPS 2021 Consistency Experiment, NeurIPS blog, Dec 2021

AI feedback led some reviewers to write more informative reviews

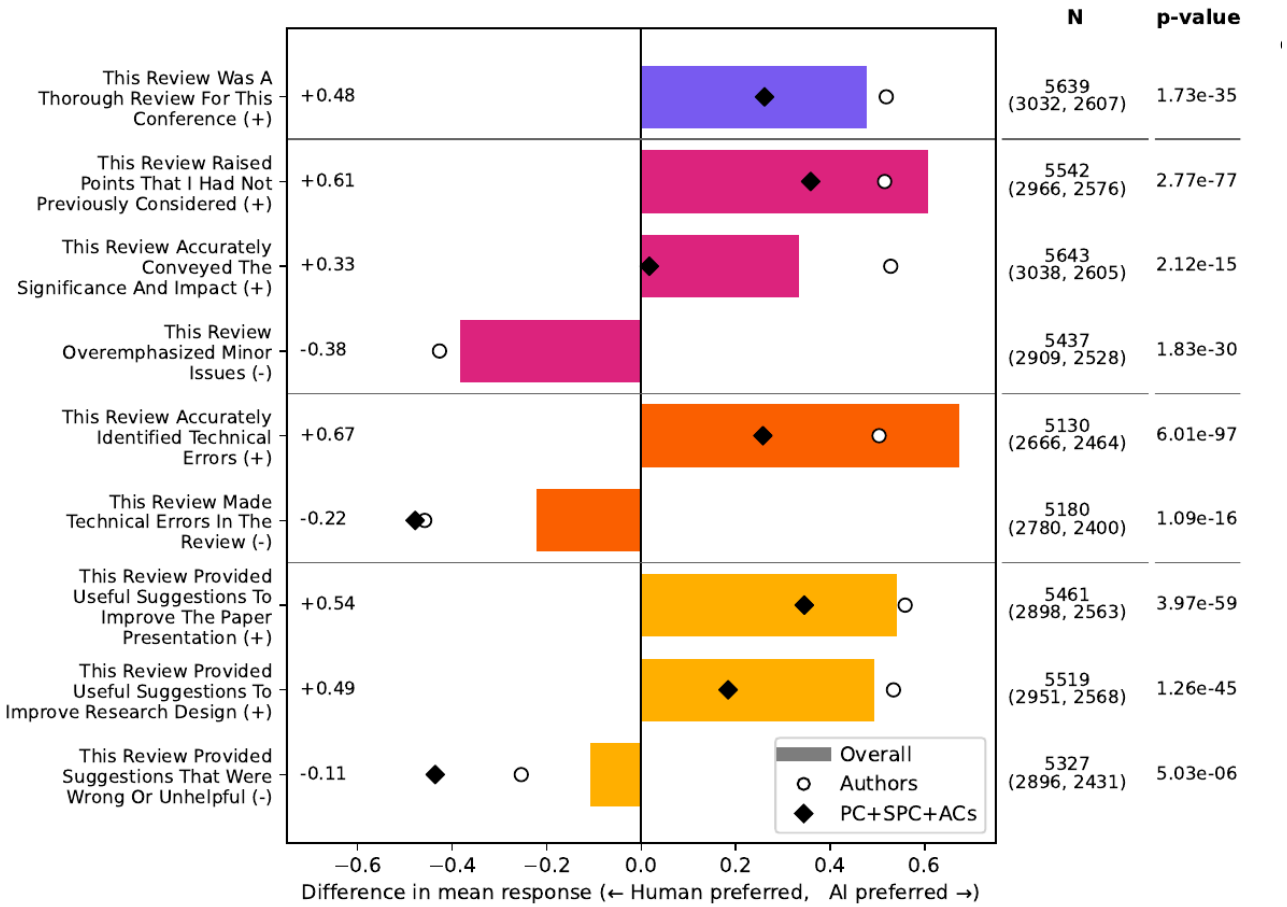
quality, with feedback only being sent to reviewers if it passed all the tests. The results show that 27% of reviewers who received feedback updated their reviews, and over 12,000 feedback suggestions from the agent were incorporated by those reviewers. This suggests that many reviewers found the AI-generated feedback sufficiently helpful to merit updating their reviews. Incorporating AI feedback led to significantly longer reviews (an average increase of 80 words among those who updated after receiving feedback) and more informative reviews, as evaluated by blinded researchers. Moreover, reviewers who were selected to receive AI feedback were also more

At ICLR 2025, an LLM agent commented on randomly chosen reviews, flagging vague or mistaken points.

Most reviewers left their review unchanged. The AI advised, and the reviewer still wrote and signed the review.

A large-scale randomized study of LLM feedback in peer review, Nature Machine Intelligence 2026

AI reviews beat human reviews on six of nine criteria



Each bar is a survey question.

Right of zero, respondents rated the AI review higher.

Human reviews won on the three negative criteria: the AI review overemphasized minor issues, made technical errors of its own, and gave some wrong suggestions.

Survey of authors and committee members. The paper notes self-selection as a caveat.

AI-Assisted Peer Review at Scale: The AAAI-26 AI Review Pilot, arXiv, Apr 2026, conference organizers

But AI reviewers' scores spread widely at Agents4Science

Our assessment of the three LLM reviewers shows that scores are positively correlated, with average pairwise Pearson correlation of 0.48. GPT-5 was the most negative reviewer, giving an average score of 2.30. In contrast, Gemini 2.5 Pro was the most positive reviewer, with an average score of 4.23. Claude 4 Sonnet was more balanced, with an average score 3.0. Among the 79 papers that were also assessed by a human expert, GPT 5 and Claude Sonnet 4 were the most aligned with the human scores, with mean absolute difference from human scores of 0.91 and 1.09, respectively. Gemini 2.5 Pro had a substantially larger mean absolute difference of 2.73.

We also found examples of sycophancy in the LLM reviews, especially from Gemini 2.5 Pro. Gemini, in one review, said, "This is a groundbreaking paper that is technically flawless, rigorously evaluated, and highly impactful. It presents a novel method that significantly advances the state-of-the-art in human-like text generation [...]". The human reviewer was more skeptical: "The literature study that reveals the several principles about human writing styles is not presented in detail, and the experiments do not show a complete analysis over the three datasets-- it looks like either the experiments on all datasets have not been completed, or only the positive results were cherry-picked for presentation."

Which LLM reviewed a paper moved its score by about two points on a six-point scale. The most lenient one also flattered.

Exploring the use of AI authors and reviewers at Agents4Science, arXiv, Nov 2025, conference organizers

ICLR's rule: disclose the LLM, and the human stays responsible

Policies on Large Language Model Usage at ICLR 2026

- Policy 1. Any use of an LLM must be disclosed, following the Code of Ethics policies that "all contributions to the research must be acknowledged" and that contributors "should expect to … receive credit for their work".
- Policy 2. ICLR authors and reviewers are ultimately responsible for their contributions, following the Code of Ethics policy that "researchers must not deliberately make false or misleading claims, fabricate or falsify data, or misrepresent results."

Using an LLM to help write a review or meta-review

the use of LLMs in their reviews. In the more extreme possibility where an LLM is used to generate a review from scratch, we highlight two potential Code of Ethics violations: First, again, the reviewer is ultimately responsible for the content of the review and consequently the reviewer would bear the consequences for LLM-generated falsehoods, hallucinations, or misrepresentations. Second, the Code of Ethics stipulates that "researchers should

Policies on Large Language Model Usage at ICLR 2026, iclr.cc

The math community wrote down the same two rules

Disclose tool use

Transparently disclose the use of automated tools, including large language models, machine learning systems, proof assistants, and other mathematical software.

Retain the responsibility for correctness

When automated techniques are employed in published mathematical research, the responsibility for the correctness and adequacy of the arguments and results, as well as for the completeness and accuracy of citations to relevant prior work, remains exclusively with the human authors.

Current automated techniques can produce plausible but unreliable (or even incorrect) arguments which are difficult to distinguish from correct mathematical proofs.

Disclose

Keep responsibility with the human authors

The risk it names first

This declaration calls for action to address the challenges posed by the use of artificial intelligence within mathematics research. It is the result of a community initiative and is endorsed by the International Mathematical Union (IMU).

Leiden Declaration on AI in Mathematics, Jun 2026

habits for judging the judge

- Ask what besides quality could sway the judge: how the answer looks, and who wrote it.
- Compare the judge with people on your own task, next to how often people agree with each other.
- Ask whether the reference can be wrong too: the answer key, the vote, the checker.
- Ask what the score rewards once it becomes a target, in training or in the prompt.
- Ask whose preferences the judge encodes, and who signs for its verdict.